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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This supplement contains the retained NBO proofs, the new centered-inference and deployment proofs, achieved readout certificates, independent validation, full-bin reconstruction and complete-work records. The unchanged preceding empirical article and supplement remain separate companions.

1. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

1.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i, let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state,

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selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \overline{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \qquad \ell \geq \log(4K/\alpha), \tag{1}$$

where K bounds the number of sampled contrasts.

THEOREM 1—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \tag{2}$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

The proof is in Supplement, Section 2.10.

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

1.2. *Replacement-fee identification*

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 2—The break-even fee frontier: *On an event where $D \in [l,u]$, the two directional gains belong to $[l,u]$ and $[-u,-l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\mathrm{dist}(0,[l,u]), \max\{|l|,|u|\}]. \tag{3}$$

For every $k \geq \max\{0,u,-l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0,l,-u\}$ only removes a certified *strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.*

The proof is in Supplement, Section 2.11.

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

1.3. *Observation costs on the same bit grid*

Let $\mathcal{B} = \{4,\dots,16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b-bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0,0]$. Let $s_b = db \sum_{t<T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 3—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b,u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \qquad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \tag{4}$$

$$\widehat{b}(\lambda) = \min \arg\min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\widehat{b}} + \lambda s_{\widehat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \tag{5}$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\widehat{b}(\lambda)$ *is weakly decreasing in* λ.

The proof is in Supplement, Section 2.12.

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (5) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr}, and its actual interpolated critic has modulus L_t^f. Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\mathrm{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \tag{6}$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\widehat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (6) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\mathrm{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

2. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

2.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

2.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

2.3. *Proof of the result 3*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$,

proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

2.4. *Proof of the result 4*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta)w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x - y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (10). *Q.E.D.*

2.5. *Proof of the result 5*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\hat{\pi}}$ is used. *Q.E.D.*

2.6. *Proof of the result 6*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the

unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

2.7. *Proof of the result 7*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r-1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$ cannot be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (15) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own

future yields the usual one-step loss recursion; backward induction gives (16). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

2.8. *Proof of the result 8*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem 7 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

2.9. *Proof of Proposition 9*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic

minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (18). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (19). *Q.E.D.*

2.10. *Proof of the result 1*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

2.11. *Proof of the result 2*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (3). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

2.12. *Proof of the result 3*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active

affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

3. PROOFS OF THE OPERATIONAL AND NEW RESULTS

3.1. *Proof of the result 10*

PROOF: Strong convexity gives $f((1-s)l + su) \leq (1-s)f(l) + sf(u) - \mu s(1-s)(u-l)^2/2$. Replacing the endpoint values by their upper bounds proves (20). Subtraction of a valid lower bound proves (21). A failed screen adds no endpoint to the partition and alters no lower bound. Subsequent protected refinement is therefore covered by the earlier termination proof. A successful screen supplies exactly the local accuracy premise used in the acquired-policy recursion. Finally, C is a convex quadratic in a, proving the sublevel-set assertion. *Q.E.D.*

3.2. *Proof of the result 11*

PROOF: A successful screen returns without an endpoint query. A failed screen invokes a procedure bounded by $K(z)$, and its margin loss equals one. This proves (24). For each fixed predictor the variable $K(z)\ell_\tau(z,\theta)$ lies in $[0, K_{\max}]$. The bounded-sum inequality of Hoeffding (1963) and a union bound over the conditionally fixed family give (25). *Q.E.D.*

3.3. *Proof of the result 12*

PROOF: The chord is a convex quadratic in its normalized action, which is affine in w. Squaring the positive part preserves convexity and gives a continuously differentiable function. The nonnegative weighted sum and quadratic penalty preserve these properties. Convexity gives $F(w) - F(v) \leq g^\top(w - v)$, and the minimum of $g^\top v$ on the box is $\sum_j \min\{0, g_j\}$. This proves (27). Every failed screen has squared positive-part residual at least τ^2, while every successful screen uses zero further endpoint queries. Summing the corresponding K_i bounds proves the work claim. Strong convexity with modulus λ proves the distance bound. *Q.E.D.*

3.4. *Proof of the result 13*

PROOF: Semiconcavity means that $f(a) - H\|a\|_2^2/2$ is concave. Its barycentric inequality, with vertex values replaced by lower endpoints, gives (28). The tangent inequality for the convex quadratic p proves (29); a linear function attains its simplex minimum at a vertex. Taking minima over a cover preserves the lower bound. Each feasible upper witness is an action of the actual model. The local gap premise therefore implies the earlier policy recursion, irrespective of how the triangle or support point was proposed.

For finiteness, use simplex lattice denominator A, with $(A+1)(A+2)/2$ vertices. On triangles of diameter proportional to $c(x)/A$, the curvature error tends to zero as A^{-2}. The endpoint error is reduced by finite descendant accuracy and quadrature refinements. The retained exact-rational recovery contract requires $\delta > 16\Lambda\, 2^{-32}$ at each recursive call, leaving positive slack after rounding; dyadic enlargement of A and q then terminates. This is a finite arithmetic procedure, not a uniform bound on rational bit cost. For (30), the independent innovation variances after conditioning on the bins are $\rho_j^2/(3q^2)$, their direction norms squared are d, and the semiconcavity Taylor remainder is $M_{t+1}/2$ times their sum. Substitution of $\rho_1 = 1/32$, $\rho_2 = 1/64$ gives the coefficient. Finally, each action query has q_r^2 descendants, proving (31). *Q.E.D.*

3.5. *Proof of the result 16*

PROOF: Conditional expectation and $b_{t+1} = b_t\beta_t$ imply

$$\mathbb{E}\sum_{t<T} b_t a_t^{\pi}(X_t^{\pi}) = \mathbb{E}\left[\sum_{t<T} b_t c_t(X_t^{\pi}, \pi_t(X_t^{\pi})) + b_T g(X_T^{\pi}) - V_0(X_0^{\pi})\right].$$

The intermediate value terms cancel. Subtracting the two expectations cancels the initial value under the common initial law, proving the identity. Summing the pointwise local bounds proves the range. Applying the bounded-sum inequality to each lower and upper endpoint sample, with failure allocation $\alpha/(2M)$, and taking a union bound proves (33). Endpoint containment supplies the comparison between endpoint expectations and the target. The width identity follows by subtraction.

No pathwise identity between D and the realized cumulative-cost difference is asserted. *Q.E.D.*

3.6. *Proof of the result 17*

PROOF: The sample-size condition implies $h_n \leq s$. Insert this and the numerical-width condition into the width identity. The final claims are set inclusions for the simultaneous confidence interval. *Q.E.D.*

3.7. *Proof of the result 18*

PROOF: The bounded-sum argument in Theorem 11 gives $Pf_\theta \leq u_\theta$ for every θ on one event of probability at least $1-\alpha$. For $0 \leq f \leq K_{\max}$, the layer-cake representation gives $|Pf - Qf| \leq K_{\max}\|P - Q\|_{\mathrm{TV}}$. Hence, on that event,

$$\mathbb{E}[N_{\theta_b}(Z_b) \mid \mathcal{F}_{b-1}] \leq \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\}.$$

The variables $c_b N_{\theta_b}(Z_b)$ have conditional ranges of length at most $c_b K_{\max}$. Conditional Hoeffding bounds, multiplied successively along the filtration and followed by the exponential Markov inequality, show that the sum exceeds its conditional means by at most the final term of (37), except on an event of probability γ. Union with the validation failure event proves the result. Every selected readout remains a proposal to the same verified gate. Its statewise certificate, and thus the earlier acquired-policy recursion, is unchanged. *Q.E.D.*

3.8. *Proof of the result 19*

PROOF: The expected cost difference in a normal block is $m\Delta$; it is zero in a declined block. Linearity of expectation gives $m(B-D)\Delta$, without requiring block independence. Multiply the cost interval by its nonnegative price, and add the work difference and fee. All price choices use the same event, so no additional probability allocation is needed. *Q.E.D.*

4. RETAINED INTERVENING TRAINING RESULTS

4.1. *Proof of Proposition 14*

PROOF: A successful screen incurs no recovery queries. On a failed screen the clipped loss equals one, and recovery work is at most WK. The pointwise bound therefore holds for the bounded variable $WK\ell_\tau$. The bounded-sum inequality and a union bound give the claim. None of the separately named service costs is included in the definition of recovery work. *Q.E.D.*

4.2. *Proof of Proposition 15*

PROOF: The lower cover still covers the entire feasible set, so its unchanged lower bound remains valid. Each additional upper endpoint bounds the true cost of its own feasible action, hence is an upper bound for the infimum. Their minimum preserves this property. The returned action's true cost is no greater than its own upper endpoint, proving the regret claim. *Q.E.D.*

5. ACHIEVED READOUTS AND INDEPENDENT VALIDATION

The two certificate-aware feature families use the same purchased states, action labels, endpoint enclosures, penalty and iteration budget. The replay verifies equality of every training-array payload at a fixed task and seed before comparing the methods. The label-loss readout shares the hat features but not the objective. The quadratic arm shares information but has a different parameter count. A finite-dimensional hat function has both a conventional piecewise-linear description and an exact ReLU realization; changing the encoding does not create a new independent method.

TABLE I: Achieved convex readout certificates

(d,T,r)	Rule	Seed	Weights	$F(w)$	$D(w)$	$F(w)/\tau^2$
2,2,2	B	1	10	0.0187	1.72e-08	7.35e+04
2,2,2	B	2	10	0.0161	2.56e-08	6.33e+04
2,2,2	H	1	10	0.0183	9.8e-08	7.19e+04
2,2,2	H	2	10	0.0154	3.26e-08	6.05e+04

(d,T,r)	Rule	Seed	Weights	$F(w)$	$D(w)$	$F(w)/\tau^2$
8,3,2	B	1	40	0.00909	2.47e-12	1.89e+04
8,3,3	B	1	40	0.0112	0	2.34e+04
8,3,2	B	2	40	0.00891	5.94e-10	1.85e+04
8,3,3	B	2	40	0.00883	1.09e-09	1.84e+04
8,3,2	H	1	40	0.00908	8.07e-10	1.89e+04
8,3,3	H	1	40	0.0112	0	2.34e+04
8,3,2	H	2	40	0.00882	8.81e-10	1.83e+04
8,3,3	H	2	40	0.00877	4.43e-10	1.82e+04

These are achieved objective and optimization-gap values, not optimizer flags or hidden-layer convergence claims. The complete rational fractions are retained in DETAILS70.json. Repeated services of the same readout are listed once here and remain separately charged in their receipts.

The objective and dual gap are evaluated on the returned quantized weights using rational contexts and features. The optimization certificate remains valid whether an approximate optimization routine reports success or a warning. The table concerns the achieved convex box problem, not unrestricted hidden-layer optimization. Raw weights, endpoints, labels and exact fractions are retained. Previously reported neural-fitting warnings belong to the unchanged earlier catalogue and are not removed by the successful computation of a new objective.

Exact context law and counting. Training and validation coordinates are uniform on the sixteen-bit dyadic grid, not on the continuous state cube or the full twenty-four-bit acquisition grid. Matching methods share these validation draws. The 6,144 method–context–date evaluations are not pooled as 6,144 independent states: every stated validation bound uses its own 256-row record. The separate cost-inference experiment retains the original continuous initial and innovation laws.

TABLE II: Independent-context work-loss validation

(d,T,r)	Rule	Seed	Returns	Root mean	Upper	All recovery queries
2,2,2	H	1	48	13.81	16.442	17680
2,2,2	H	2	50	13.68	16.315	17510

(d,T,r)	Rule	Seed	Returns	Root mean	Upper	All recovery queries
2,2,2	B	1	25	15.34	17.000	19635
2,2,2	B	2	21	15.61	17.000	19975
2,2,2	L	1	42	14.21	16.973	18190
2,2,2	L	2	47	13.88	16.482	17765
2,2,2	Q	1	45	14.01	16.707	17935
2,2,2	Q	2	52	13.55	16.282	17340
8,3,2	H	1	21	15.61	17.000	19975
8,3,3	H	1	23	15.47	17.000	462088
8,3,2	H	2	13	16.14	17.000	20655
8,3,3	H	2	21	15.61	17.000	469565
8,3,2	B	1	25	15.34	17.000	19635
8,3,3	B	1	23	15.47	17.000	462088
8,3,2	B	2	11	16.27	17.000	20825
8,3,3	B	2	20	15.67	17.000	471130
8,3,2	L	1	9	16.40	17.000	20995
8,3,3	L	1	6	16.60	17.000	486452
8,3,2	L	2	5	16.67	17.000	21335
8,3,3	L	2	13	16.14	17.000	481419
8,3,2	Q	1	8	16.47	17.000	21080
8,3,3	Q	1	21	15.61	17.000	464894
8,3,2	Q	2	10	16.34	17.000	20910
8,3,3	Q	2	4	16.73	17.000	493938

Each row has 256 new contexts; seed suffix 1 or 2 denotes 6901 or 6902. The upper bound is independently recomputed from an exact sample sum and an outward radius. It applies to the declared complete-action mesh recovery and context law, not automatically to an adaptive on-policy stream. Recovery-query totals include descendants.

The validation family contains twenty-four task–date–method–seed records with 256 contexts per record. The logarithm allowance twelve covers the stated finite family by a union bound. Reusing a context stream across methods induces dependence between estimators but does not invalidate that bound. The same stream is not counted as additional independent information for each method. The deterministic

work inequality is checked on every context. The distributional upper endpoint is recomputed from the exact sample sum, an eighty-digit square root with an explicit expansion, and outward conversion to the printed numerical endpoint.

The full-action recovery mesh in this validation design has a declared root cap K. Its actual descendant queries are separately recorded. These observations do not validate a different adaptive fallback by definition. In Theorem 18, a distributional transfer allowance and a complete endpoint-work cap are additional premises. The theorem therefore handles predictable choices and dependent contexts without asserting that the recorded recursive state stream is independent.

6. PRECISION AND THE CONTINUOUS-LAW ESTIMAND

TABLE III

PAID INFERENCE WORK AND RAW ENCLOSURE COMPONENTS

(d, T)	Rule	Evaluator queries	Arm (s)	Conditional	State-bin	Ambiguous
2,2	A	101226	1.98	0.000113	2.3e-13	0
2,2	B	101226	1.91	0.000113	2.3e-13	0
2,2	H	101226	1.91	0.000113	2.3e-13	0
2,2	R	101226	1.83	0.000113	2.3e-13	0
8,3	A	2921243	100.74	0.000595	3.7e-13	0
8,3	B	2921234	102.56	0.000596	3.7e-13	0
8,3	H	2921234	102.59	0.000596	3.7e-13	0
8,3	R	2921465	101.36	0.000597	3.7e-13	0

Component columns are mean full widths before valid regret-range intersections. They do not add to the final narrower contrast width. Arm clocks are internal diagnostics; complete shared inference is charged as a whole: 9.03 seconds, 408.63 seconds. State-bin widths include propagated arithmetic. No acquisition ambiguity is discarded.

The centered observation is the sum of optimal Bellman advantages along an actual policy trajectory. Its expectation equals policy loss. Subtracting two such observations cancels the common initial optimal value in expectation. It does not cancel cash-flow risk path by path, and no empirical fitted value is substituted

for the optimal value. Each conditional Bellman evaluation uses its own tolerance, action covering and continuous-innovation integration account, whose work is charged to the inference service.

The twelve paired intervals form one finite family over the two declared scalar tasks. With logarithm allowance twelve, $2 \cdot 12 \exp(-12) < 0.01$. For example, a rational partial sum of the exponential series already proves $\exp(12) > 2400$; no rounded logarithm is needed for the coverage claim. The ordinary cash-cost estimator uses a separate empirical-Bernstein family on the same rows. A joint claim about both families requires the sum of their tail budgets, not an assumption that the estimators are independent.

The mean conditional-value and state-bin component widths are recorded before intersecting with valid local regret ranges. The intersection can substantially reduce the resulting interval. Accordingly, those diagnostic widths need not sum to the final centered width, and they are not interpreted as independent errors. The state-bin column includes propagated state arithmetic. The actual final interval width is the mean final numerical width plus the two sampling radii, before any final support clipping.

For every path, the numerical action used by the controller is fixed before the more accurate evaluation. Its feasibility is checked on the true-state box as well as on the acquired state. An unresolved acquisition branch must receive a valid remaining support; it cannot be discarded. None is encountered in the new inference catalogue. The absence of sampled ambiguity is an observation, not a theorem excluding acquisition boundaries from the model.

7. INDEPENDENT FULL-BIN RECONSTRUCTION

The new independent implementation imports no NBO numerical kernel. It uses exact rational intervals for the continuous initial and innovation bins, twenty-four-bit acquired-state comparisons, true-state feasibility, stage costs, transitions and terminal costs. It treats the two cumulative endpoints as separate immutable rational numbers. In a transition coordinate, the deterministic function

$$f(u,v) = \frac{1}{16} + \frac{u}{2} + \frac{v}{8} + \frac{u(1-v)}{16}$$

has derivatives $1/2 + (1 - v)/16 \geq 0$ and $1/8 - u/16 \geq 0$ on the state cube. Thus evaluation at the lower–lower and upper–upper corners bounds its range without invoking the production interval library. Quadratic costs use exact interval squares; shortage uses the monotone positive-part map followed by an interval square. Both shocks keep their original widths and marginal laws.

The implementation checks that its complete bin enclosure lies inside the archived cash-cost enclosure. A second, independently written rational midpoint calculation checks acquisition and cumulative cost at a represented point. The midpoint test is not substituted for the full-bin test. Across 3,706 physical path archives there are 38,722 recorded path rows; 736 distinct payloads contain 14,242 unique rows. Repeated timing copies are hash-checked and compared, but their duplicate rational evaluation is not counted again. The independent full-bin replay covers all 14,242 unique rows, with no unresolved branch in this executed catalogue.

A separate exact-sum and high-precision calculation checks every final centered confidence interval from its stored advantage endpoints. This independently verifies final inference arithmetic, not the generation of those optimal-advantage endpoints. The regression suite compares primitive identities and small complete value problems with separate rational formulas. It does not independently regenerate the largest multivariate Bellman descendant tree. The release states these different coverage levels rather than assigning an undifferentiated independent-verification label to the whole program.

7.1. *Prediction portability and retained disagreements*

An initial strict replay stopped because one Bernstein validation prediction was not bitwise equal to a recomputed value. That failed attempt, its original audit source and traceback are retained on their own branch and in the publication inputs. The failure does not by itself identify an invalid Bellman bound. Nor does a successful rerun on another processor prove the earlier numerical result bitwise reproducible.

The current audit checks the saved actual action on its declared lattice and recomputes its strong-convexity chord with exact rational arithmetic. A discrepancy with

a recomputed readout is additionally checked against the declared action-rounding and prediction allowances, not accepted by an arbitrary numerical closeness threshold. The original scientific endpoints, actions, models and source freeze are unchanged. Numerical parameter identity, callable-policy identity and successful certificate verification are distinct statements.

In particular, equality of saved readout arrays is necessary but not sufficient for transferring a cost estimate to an implementation on a different arithmetic runtime. The complete-service purchase region assumes the same numerical callable policy or an additional valid implementation-transfer bound. Matching processor metadata or passing a finite replay is not silently promoted to an all-state policy-identity theorem.

8. COMPLETE WORK AND CONDITIONAL ADOPTION

TABLE IV: Conditional reuse-stage adoption thresholds

(d, T)	Rule	Repeat	Saving (s)	Loss upper	p/v threshold
2,2	H	0	0.000	0.3441	1383.7120
2,2	H	1	0.350	0.3441	0.9821
2,2	H	2	-0.201	0.3441	–
2,2	H	3	3.457	0.3441	0.0996
2,2	B	0	-0.201	0.3440	–
2,2	B	1	0.601	0.3440	0.5727
2,2	B	2	-0.401	0.3440	–
2,2	B	3	3.808	0.3440	0.0904
8,3	H	0	-1.153	0.8079	–
8,3	H	1	-1.301	0.8079	–
8,3	H	2	-1.399	0.8079	–
8,3	H	3	-1.555	0.8079	–
8,3	B	0	-1.306	0.8079	–
8,3	B	1	-1.606	0.8079	–
8,3	B	2	-1.302	0.8079	–
8,3	B	3	-1.955	0.8079	–

Seed 6901; zero incremental installation fee and common or previously paid qualification. The loss upper is 504 times the corresponding policy-cost upper endpoint. A displayed positive-saving threshold describes the adoption region only when the numerical callable policy is the one evaluated by inference. Saved parameter equality is checked, but cross-platform numerical equality is an additional requirement. A dash means no positive work saving in that matched observation. Newly purchased qualification work must be added before claiming a first-purchase saving. Prices are declared inputs, not estimated currency values.

Qualification is not free. The displayed reuse thresholds condition on a common or previously paid qualification calculation. The process receipts cover the complete reuse service, not the separate policy-cost inference experiment. For a first purchase, let C_q be additional qualification work not paid by the benchmark alternative. In Proposition 19, use $W_\pi = W_\pi^{\mathrm{run}} + C_q$ and $W_\nu = W_\nu^{\mathrm{run}}$; the observed saving becomes $S - C_q$. This is a change of the explicit accounting boundary, not an additional theorem assumption about free inference. Charging the entire observed shared inference process gives C_q approximately 9.03 seconds in the small task and 408.63 seconds in the larger task. Every displayed matched saving then becomes negative. Thus the current receipts do not identify a positive qualification-plus-reuse work saving under that conservative first-purchase boundary. An allocated marginal inference time or a longer repayment horizon was not separately executed. The machine-readable table retains both boundaries, and the general price region remains valid with the actual charged work and installation fee.

The multiplier 504 follows from sixty-four blocks of eight trajectories with one predetermined fallback block. In that block the service uses the benchmark itself, giving zero expected policy-cost difference. In the other blocks it uses the fixed candidate. The cost interval can therefore be scaled by 504 under the policy-identity premise of Proposition 19. A positive work saving is an observed matched process difference, not a confidence bound for future latency. The complete timing range also records all unfavorable repetitions and does not authorize selection of a favorable timing result as a population estimate.

The parent receipt includes child launch, imports, fitting or loading, all policy and path calculations, serialization, child exit and log flushing. Parent receipt

bookkeeping belongs to the group clock. Training, inference and publication replay have different boundaries and are not assigned zero cost when excluded from a displayed sub-account. The machine record includes the permitted CPU, thread counts, available frequency and governor data. Processor frequency was not locked by the experiment. Fine timing differences consequently remain descriptive even when operation counts reproduce exactly.

The original normalized investment units are retained. The loss margin is fixed by the numerical protocol; the value of a loss unit, compute price and installation fee are purchaser inputs. This produces a transparent conditional adoption region, not an estimated monetary welfare result. Its noninferiority and equivalence statements concern the actual expected-cost target and do not imply strict neural superiority.

9. PRESERVATION AND SCOPE

Every active R67 theorem statement is retained verbatim in the current theoretical sections, and every proof is retained in the article or this supplement. The original empirical article and its full supplement are unchanged companion documents. They retain all earlier conventional winners, unresolved direct-cost comparisons, fitting warnings, vector horizon costs and the accumulation-aliasing counterexample. The broader NBO development remains available under its original diffusion, recursive-preference, temporal-self and game hypotheses; those hypotheses are not replaced by the convex expected-cost premises of the current query theorem.

The new frozen R69 catalogue is integrated once. Publication replay, repeated clocks, repeated random streams, duplicate path payloads and regenerated tables do not increase its number of independent economic observations. The final source and content map records the location and hash of each retained item. Compilation, regression tests and independent numerical calculations are evidence for their stated checks, not an external editorial decision or a formal proof-checker certificate.

10. COMPLETE COLD-START AND VECTOR WORKLOADS

The tables retain every declared method and task. They are relocated intact from the article to keep its reading path focused; no row, adverse comparison or timing repetition is removed.

TABLE V

COMPLETE COLD-START SERVICES AT A COMMON INFORMATION BUDGET

(d, T)	Rule	Processes	Policy queries	Median (s)	Range (s)
2,2	A	4	597	0.365	[0.364, 0.366]
2,2	B	8	607–612	0.465	[0.465, 0.466]
2,2	H	8	592–612	0.465	[0.465, 0.466]
2,2	L	8	592–597	0.440	[0.414, 0.466]
2,2	Q	8	597	0.415	[0.415, 0.967]
2,2	U	4	597	0.390	[0.364, 0.415]
8,3	A	4	2594	0.666	[0.615, 0.666]
8,3	B	8	2594	1.618	[1.618, 1.619]
8,3	H	8	2594	1.618	[1.618, 1.669]
8,3	L	8	2594	1.518	[1.517, 1.568]
8,3	Q	8	2594	1.518	[1.467, 1.519]
8,3	U	4	2594	0.641	[0.615, 0.666]

A: adaptive; U: upper-chord minimizer; H: certificate-aware hat–ReLU; B: certificate-aware Bernstein; L: label-loss hat–ReLU; Q: quadratic. Queries concern the policy workload; training and validation queries are additional. Four process repetitions per seed are descriptive clocks, not independent fits or confidence bounds. All complete times include launch and shutdown.

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TABLE VI

TWO-CONTROL AND TWO-INNOVATION COMPLETE SERVICES

(d, T)	Rule	Processes	Queries	Median (s)	Peak KiB
2,2	H	2	98	0.47	65520
2,2	Q	2	99	0.49	65524
2,2	S	2	82	0.39	65164
2,2	U	2	82	0.36	65328
2,3	H	2	5634	5.83	66004
2,3	Q	2	5914	5.93	66256
2,3	S	2	4541	3.70	65184
2,3	U	2	9039	7.93	65016
2,4	H	1	872033	62.01	65872
2,4	S	1	592658	41.37	65256
8,2	H	2	91	0.46	65780
8,2	Q	2	91	0.46	65864
8,2	S	2	75	0.36	65252
8,2	U	2	75	0.36	65336
8,3	H	2	3495	5.93	66028
8,3	Q	2	3495	5.85	66208
8,3	S	2	2645	3.55	65168
8,3	U	2	3278	4.60	65544

S: simplicial; U: largest-edge uniform refinement; H and Q: trained two-output hat and quadratic proposals. Ordinary services have eight continuous-bin paths. Horizon-four services have one path, and H transfers its three-date readout. These are workload records, not precision-designed estimates of population policy costs. Peak memory is the maximum observed resident set within the listed processes.